



PATH 01 / NEW TERRITORY / PROVINCE-LEVEL PRELIMINARY SCREENING

Territory Briefing: Torino

Evidence-based preliminary infrastructure intelligence for technical meetings, early design discussion and due diligence. ARCUS supports screening; it does not replace site-specific checks or design verification.

REPORT ID

**ARCUS-P01-
20260609-torino**

**PROJECT
CONTEXT**

Bridge

DOMINANT DRIVER

Hydraulic

EVENTS

42

SPATIAL LEVEL

Province level

ANALYSIS TYPE

Territory Briefing

VERSION

**ARCUS Professional
v1.0-preview**

GENERATED BY

ARCUS Professional

Executive Summary

ARCUS is based on the first systematic database of collapsed bridges in Italy from 2000 to today. The value of this report is not map aggregation: it is the translation of documented collapse evidence into operational infrastructure intelligence.

ARCUS identifies 42 documented cases in the selected province and translates the Step 3 exposure indicators into operational actions for Bridge interventions. The PDF map shows ARCUS cases; the hazard-layer contribution is summarised through the exposure indicators below. The dominant territorial signal is **Hydraulic**; the leading historical cause is **Hydraulic**.

Recommended use: province-level preliminary screening before design, due diligence or site-specific investigation.

Dominant Exposure
Hydraulic / score 97

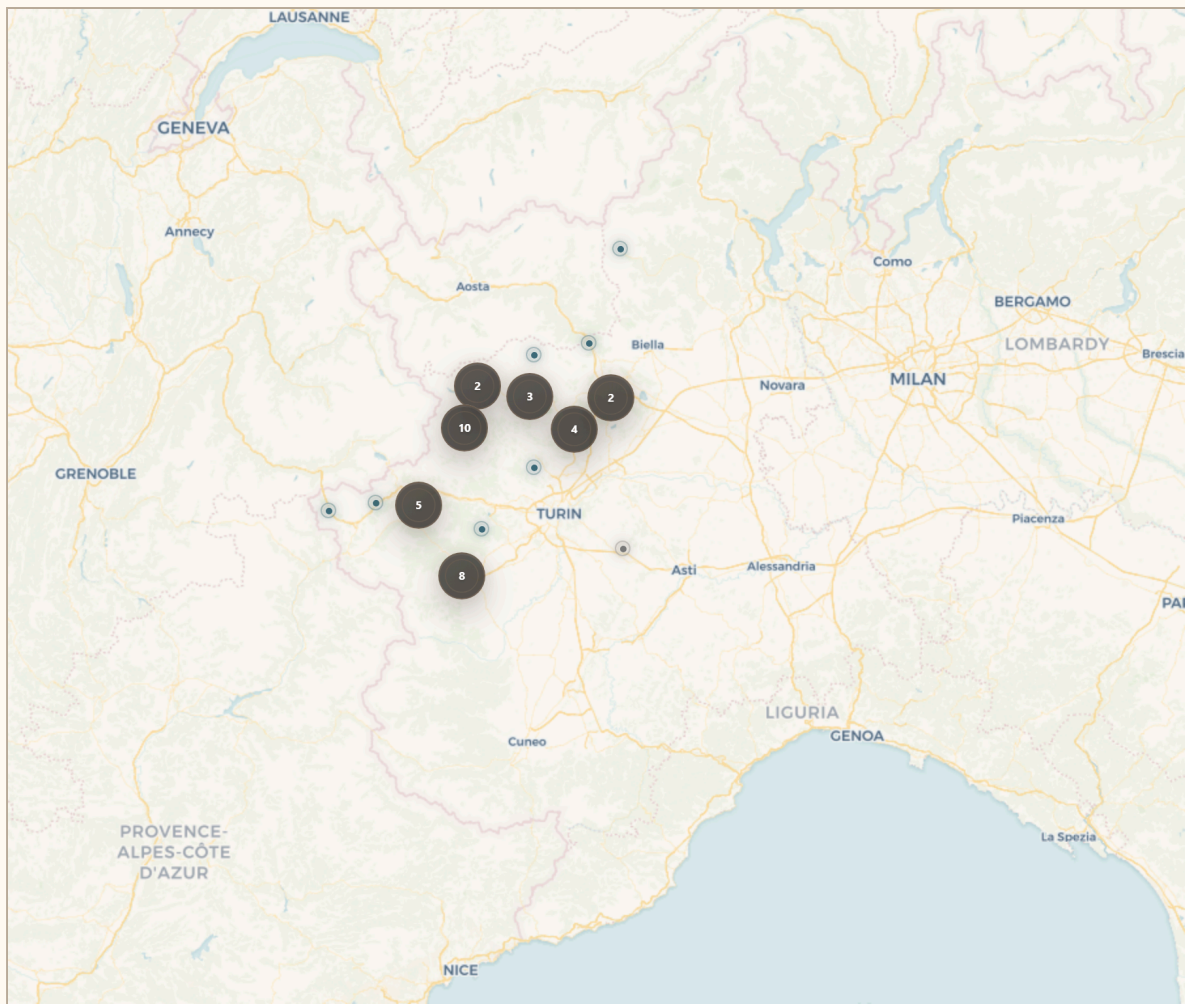
Historical Signal
Hydraulic represents 98% of the selected evidence.

Priority Use
Verify whether the planned crossing interacts with active riverbeds or flood-prone corridors.

WHY ARCUS FLAGS THIS

- ARCUS is based on the first systematic database of collapsed bridges in Italy from 2000 to today.
- Historical severity: 30 total collapses and 41 triggered events in the selected province.
- Map-linked cases: P1 Castellamonte, P2 Ala di Stura, P3 Ala di Stura.
- Dominant exposure: Hydraulic (97 / 100).
- Evidence window: 2000-2024 (25 years of evidence).
- Evidence package: 125 linked sources supporting the screening signal.

ARCUS Cases Map



Static ARCUS cases map - full interactive view available in Professional Atlas.

How to Read

ARCUS starts from the selected province, reads all ARCUS records available for that administrative area, identifies the dominant territorial pattern and turns that signal into operational next checks. This is province-level screening, not structural safety certification.

Province-level reading is the minimum comparable screening scale. Large provinces can include mountain valleys, plains, hills and urban corridors; a specific site, municipality or alignment requires a more granular follow-up analysis.

SELECTED PROJECT CONTEXT

Bridge

Major bridge or bridge intervention: river crossing behaviour, hydraulic compatibility, scour, foundations, piers, abutments, structural concept and inspections.

DOMINANT EXPOSURE

Hydraulic

41 matched ARCUS events. Public overlays are used as screening context and must be followed by site-specific checks.

Key Indicators

ARCUS EVENTS

42

Selected province

TOTAL COLLAPSE

30

Historical severity

TRIGGERED EVENTS

41

External initiating
condition

EVIDENCE WINDOW

2000-2024

2000-2024 (25 years of
evidence)

Evidence package: 125 linked and deduplicated sources support the selected provincial screening signal.

LAYER 1 / TERRITORIAL EXPOSURE

Priority Index

82 / 100

Measures exposure and hazard context. It answers:
this territory has elevated hazard/exposure
characteristics.

LAYER 2 / HERITAGE VULNERABILITY

Collapse Rate

5.82x

3.268 ARCUS cases per 100 AINOP bridges;
denominator 1285 AINOP bridges.

CONFIDENCE: HIGH

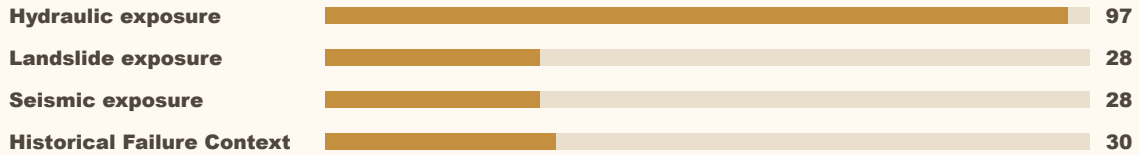
Read the two indicators as separate layers, not as a composite score. A province can show high territorial exposure with a low Collapse Rate, or low exposure with a high Collapse Rate; the combination is more informative than a single merged number.

ARCUS/AINOP Collapse Rate: 3.268 ARCUS cases per 100 AINOP bridges; 5.82x the national ARCUS/AINOP rate. Denominator: 1285 AINOP bridges (1285 road, 0 rail). Confidence note: AINOP denominator is broad enough for provincial relative benchmarking.

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Hazard / Exposure Context

The Step 3 analysis combines public Hydraulic, Landslide and Seismic layers with ARCUS historical cases. For PDF readability, the map displays ARCUS cases only. The influence of the active hazard layers is represented through the exposure scores below.



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Operational Meaning

ARCUS READING

ARCUS reading: For bridge interventions in the selected province, ARCUS indicates that preliminary attention should focus on hydraulic compatibility and related checks because the dominant indicator is hydraulic exposure. This signal does not define a design solution. It identifies the technical attention fields that should be checked before design, due diligence or site-specific investigation decisions.

PRIMARY FOCUS

Hydraulic compatibility; scour susceptibility; foundation, pier and abutment exposure.

ATTENTION POINTS

Hydraulic compatibility

Scour susceptibility

Foundation, pier and abutment exposure

Riverbed dynamics and sediment/debris transport

Flood/debris scenarios

Inspection and monitoring accessibility

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Priority ARCUS Cases

The selected province contains all ARCUS records available for that administrative area. This section highlights the highest-priority cases within the provincial set, ranked by severity, trigger, dominant driver and evidence reliability. These cases are not external comparables: they explain the dominant territorial signal used by the report.

Note: P1-P3 refer to the same 13-16 October 2000 flood event in Piemonte and Valle d'Aosta. This contextualises the signal as documented extreme-flood vulnerability, not three independent recurrent events.

P1

B00.10.15 - Castellamonte 2000-10-13 / Hydraulic

TC / High

P2

B00.10.03 - Ala di Stura 2000-10-13 / Hydraulic

TC / High

P3

B00.10.05 - Ala di Stura 2000-10-13 / Hydraulic

TC / High

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Territorial Pattern

The selected province shows a concentrated Hydraulic pattern. This pattern supports preliminary attention toward hydraulic compatibility; scour susceptibility; foundation, pier and abutment exposure. for Bridge interventions.

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Interpretation

For Bridge interventions in Torino, ARCUS indicates that preliminary investigation should be framed around Hydraulic, documented cases and the operational consequences implied by the selected project context.

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Immediate Actions

ACTION 1

Verify whether the planned crossing interacts with active riverbeds or flood-prone corridors.

ACTION 2

Review hydraulic-triggered total-collapse precedents in the selected province.

ACTION 3

Prioritise hydraulic modelling and scour screening before preliminary design consolidation.

ACTION 4

Check foundation, pier and abutment exposure to riverbed evolution and debris accumulation.

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Data to Request

DATA 1

Hydraulic model or preliminary hydraulic study

DATA 2

Riverbed survey / bathymetry where relevant

DATA 3

Historical flood levels

DATA 4

Geotechnical investigation

DATA 5

Foundation concept or available foundation records

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Design-Phase Checks

ACTION 1

Translate province-level screening into site-specific checks by qualified professionals.

ACTION 2

Review P1-P3 priority ARCUS cases as evidence of the province-level signal.

ACTION 3

Use exposure indicators as a synthesis of public Hydraulic, Landslide and Seismic layers, not as detailed hydraulic, geotechnical or seismic modelling.

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Recommended Client Path

1. Use the Step 3 exposure indicators to identify the dominant Hydraulic signal.
2. Review P1-P3 Priority ARCUS Cases.
3. Request the minimum technical data package.
4. Commission site-specific hydraulic and geotechnical checks.
5. Use CSV and GeoJSON exports for technical coordination.

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Data Coverage & Limitations

Province-level screening only; not design scale.

ARCUS does not certify structural safety or asset condition.

Public overlays support prioritisation and must be followed by site-specific investigations.

ARCUS currently reports the provincial evidence window as 2000-2024 (25 years of evidence). The Collapse Rate is a derived ARCUS/AINOP indicator: AINOP is fed by owners/managers and may under-represent local assets, especially in small mountain municipalities. For this province the denominator confidence is high: AINOP denominator is broad enough for provincial relative benchmarking.

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Methodology Snapshot

ARCUS combines documented collapse events, source reliability, spatial relevance, severity, triggers, cause distribution, territorial hazard context and province-level priority cases. The dominant exposure indicator is calculated from Step 3 scores: hydraulic, landslide, seismic and historical failure context.

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Score and Class Legend

P1/P2/P3 identify the highest-priority ARCUS cases within the selected province on the exported Atlas map. Severity TC means total collapse; triggered events indicate an external initiating condition was recorded. Use "Historical Failure Context" for province-level screening unless asset-level structural data are available.

A

Short Source Appendix

B24.09.01 / PROTEZIONE CIVILE

Evento meteo-idrologico 05 settembre 2024

B17.06.01 / JOURNAL PAPER

Bridge collapses in Italy across the 21st century:
survey and statistical analysis

B08.05.02 / JOURNAL PAPER

Bridge collapses in Italy across the 21st century:
survey and statistical analysis

B08.05.02 / REGIONE PIEMONTE

Evento alluvionale 29-30 maggio 2008 – relazione
tecnica ufficiale

B05.11.01 / JOURNAL PAPER

Bridge collapses in Italy across the 21st century:
survey and statistical analysis

B00.10.03 / ARPA PIEMONTE

Evento alluvionale regionale 13-16 ottobre 2000 –
monografia ufficiale

B00.10.03 / JOURNAL PAPER

Bridge collapses in Italy across the 21st century:
survey and statistical analysis

B00.10.04 / ARPA PIEMONTE

Evento alluvionale regionale 13-16 ottobre 2000 –
monografia ufficiale

The full source table is provided as a separate supporting CSV export.